TECLA WAIRIMU GICHEHA

NETWORKING • NETWORK SECURITY • IT INFRASTRUCTURE

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CAREER SUMMARY

CCNA Certified Network Engineer specializing in enterprise routing, network security, and infrastructure support. I have Experience in configuring and maintaining routed and switched networks, managing Network Security policies, and developing Python automation scripts to eliminate manual configuration errors. I have hands on experience in implementing secure network architectures and troubleshooting IT infrastructure issues to keep systems running smoothly.

WORK EXPERIENCE

Ministry of Trade and Industry (State Department Of Trade) – Telposta Towers, Kenya

Network and IT infrastructure Intern

- **Optimized Local Infrastructure** by actively configuring and maintaining core office networks, ensuring seamless connectivity across departmental branches
- **Reduced Network Downtime** by continuosly monitoring hardware health, verifying port configuration and resolving local connectivity faults
- **Hardened Enterprise Security** through deployment of robust access (ACLs) and strict compliance audits across multiple end-user endpoints

Amref Health Africa – Lang'ata, Kenya

Network and IT infrastructure Intern

- **Maintaining System Availability** by executing routine infrastructure health checks, resolving help desk ticket and providing Technical Support
- **Facilitated Technical Logistics For High-level hybrid meetings** and Regional Webinars ensuring stable audiovisual streams and secure remote connections
- **Reduced Network Downtime** by continuosly monitoring hardware health, verifying port configuration and resolving local connectivity faults
- **Conducted Physical And Logical Network Troubleshooting** using basic utilities like Ping,Traceroute and NSLookup to isolate diagnose, resolve day-to-day client connectivity faults

SKILLS

Networking:	TCP/IP, IPv4 subnetting, VLANs, trunking, inter-VLAN routing, OSPF, EIGRP, routing fundamentals.
Network Security:	Access Control Lists (ACLs), DMZ setup, IPsec VPN, AAA/RADIUS/TACACS+, switch port security, DHCP snooping, DAI.
Infrastructure & Tools:	GNS3, Cisco Packet Tracer, VMware Workstation, GNS3 VM, Ubuntu, Wireshark, GitHub.
Automation & Scripting:	Python fundamentals, Netmiko, Jinja2, YAML templating, SSH automation workflows.

PROJECTS

Enterprise Infrastructure Security Architecture

Built and secured a resilient, multi-site enterprise network environment within Cisco Packet Tracer to simulate real-world corporate connectivity and perimeter defenses. To protect internal corporate data, I isolated public-facing resources inside a dedicated DMZ using an edge gateway configured with strict, unidirectional Extended Access Control Lists (ACLs) to completely block lateral traffic from moving into the private LAN. I engineered a secure site-to-site IPsec VPN tunnel across a simulated ISP using robust AES-256 encryption and SHA-256 hashing to bridge the branch office securely to headquarters. To eliminate internal insider threats and unauthorized access, I deployed centralized TACACS+/RADIUS authentication for the routing infrastructure and hardened the access layer against man-in-the-middle attacks by implementing port security with MAC-address lockdown, DHCP snooping, and Dynamic ARP Inspection.

Network Building Automation

Engineered a scalable network automation workspace combining GNS3, VMware Workstation, and an Ubuntu development node to automate device provisioning and lifecycle management. I developed modular Python scripts using the Netmiko library to handle concurrent, multi-threaded SSH connections, allowing configuration payloads to deploy to multiple virtual network endpoints simultaneously. By decoupling configuration logic from device data, I structured all environmental variables inside clean YAML files and mapped them onto Jinja2 templates, generating consistent, vendor-agnostic baseline configurations. This programmatic pipeline eliminated human error during manual CLI setups, cut configuration deployment timelines drastically, and established a reliable source of truth across all managed network devices.

CERTIFICATES

Cisco Certified Network Associate (CCNA v7)

CiscoNetworkingAcademy • (Offered by Jomo Kenyatta University) • Active

Completed enterprise-level training centered on routing and switching architectures, core network infrastructure protocols, and secure system implementations. Developed deep technical competency in provisioning, securing, and troubleshooting complex corporate network topologies. Gained foundational, hands-on experience in modern network programmability, utilizing REST APIs, and leveraging automated configuration management tools to design and support next-generation software-defined networking (SDN) environments.

Network Security Certificate

CiscoNetworkingAcademy (Offered by African Advanced Level Telecommunications Institute) •Active

Completed specialized training in architecting secure perimeter defenses and mitigating advanced threats across distributed enterprise infrastructures. Hands-on experience includes implementing secure firewall zones, configuring multi-site cryptographic IPsec VPN paths, and deploying centralized AAA authentication frameworks. Competent in locking down network access layers using port security validation mechanisms and advanced monitoring systems to maintain end-to-end infrastructure integrity.

EDUCATION

Multimedia University of Kenya

Bachelor of Science in Information Technology

September 2022 – July 2026